Spatial–Spectral Adaptive Learning With Pixelwise Filtering for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification is significant in remote sensing applications. However, most methods focus on the spectral and spatial correlation information in the neighborhood while ignoring the feature difference among global different pixels. In this article, we propose a spatial–spectral adaptive learning with pixelwise filtering (SSALPF) method to fully consider the discriminative information of pixels in different spatial locations, which mainly consists of a parallel spatial–spectral adaptive learning (SSAL) module and a pixelwise filtering (PF) module. Specifically, the former aims to obtain joint spatial–spectral discriminative features of each pixel point in a parallel manner and is used as a guide for adaptive selection of filter kernel. The latter uses the adaptive filter kernel to implement pixel-level filtering on HSI, in order to learn the discriminative features contained in different pixel points for classification. The adaptive filter kernel is generated by a linear combination of a predefined dictionary containing multiple filter bases. Experiments demonstrate that the proposed method is superior to other methods on popular hyperspectral datasets.**

***Index Terms*— Adaptive learning, adaptive selection kernel, feature extraction, hyperspectral image (HSI) classification, pixelwise filtering (PF).**

I. Introduction

HYPERSPECTRAL image (HSI) receives great interest in remote sensing because of its rich spatial information and spectral information with hundreds of contiguous narrow bands [1]. Thus, HSI contains information that can be used to represent the distribution of objects and to distinguish between categories. In recent years, hyperspectral technologies are widely applied in a variety of fields, such as Earth observation [2], [3], resource management [4], chemical imaging [5], and environmental monitoring [6], [7]. The aim of HSI classification is to predict the category for each pixel point [8].

In the past, a large number of methods use handcrafted features to achieve the HSI classification [9], [10], such as support vector machine (SVM) [11], subspace [12], and linear or logistic regression [13], [14]. During the exploration of feature extraction, some methods of dimensionality reduction and feature selection are also applied in the HSI classification. In the dimensionality reduction approach, principal component analysis (PCA) replaces the original spectral vector with a reduced-dimensional spectral vector for classification [15]. The random forest (RF) method is also employed for the feature selection process of the classification task [16]. Another independent component analysis (ICA) expresses the HSI characteristics by assuming a linear combination of independent components [17]. However, traditional feature extraction methods are designed based on the data characteristics themselves and therefore do not match different types of data, while deep learning models trained in an end-to-end way are able to learn the features of different types of input data and output predictions to complete the HSI classification task. Autoencoder (AE) explores reconstructed features for HSI classification [18]. Stacked AE (SAE) increases the depth compared to AE, thus enhancing the feature expression of HSI classification [19]. The convolutional layer of convolutional neural network (CNN) is also available for obtaining deep features [20]. In contrast to CNN, recurrent neural network (RNN) considers the effect of features of time series on HSI classification results [21], [22]. Generative adversarial network (GAN) for HSI classification is designed in an unsupervised generative adversarial manner [23].

In the early stage, attention is focused on the spectral information of HSI, where a large number of methods based on pixel point vectors are proposed for HSI classification. In particular, some of the traditional classification methods mentioned above exploit spectral vectors, such as SVM [11], PCA [15], and ICA [17]. Besides, the original spectral vector of HSI is also used as an input for training SAE models in deep learning studies [1]. Some methods for 1-D spectral vector, such as 1-D CNN and 1-D RNN, are used to extract HSI spectral features [24], [25]. However, there are studies showing that considering spatial information can further improve the

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classification accuracy [26], [27]. The k-nearest neighbor (KNN)-based spectral–spatial feature fusion method has good performance on HSI classification [26]. HSI classification accuracy can also be advanced by utilizing SVM to encode the spatial and spectral information of the pixels [27]. The HSI vector after PCA dimensionality reduction is used as the input of 2-D CNN to extract the spatial information contained in the spatial neighborhood [28]. In contrast to [28], the spatial feature extraction network is replaced by AlexNet in [29], while the balanced local discriminant embedding (BLDE) of [30] substitutes the simple PCA to achieve spectral feature extraction and finally fuses the two features for classification. In addition, the emergence of methods for joint extraction of spatial–spectral features that more fully exploit the rich properties of the original HSI [31], [32]. The 3-D depth capsule network (CapsNet) approach using vectors rather than scalars enhances the representation of spectral–spatial features [33]. Unsupervised cooperative sparse AE (CSAE) obtains discriminative features for HSI classification by fusing multiscale spectral–space features [34].

Recently, attention mechanism is popular in HSI classification tasks due to its ability to focus attention on certain aspects of interest [35]. Therefore, combining attentional mechanism with CNN-based models already has been displayed better performance [36], [37]. Squeeze-and-excitation network (SENet) improves network performance by modeling channel relationships with attention [38]. In SENet, the dimensional decay of the fully connected (FC) layer affects the weight learning of channel attention, which in turn affects the correct classification of HSI. Efficient channel attention network (ECA-Net) utilizes adaptive 1-D convolution instead of FC layer to achieve cross-channel information interaction and outperforms SENet [39]. The convolutional block attention module (CBAM) considers acting both spatial attention and channel attention into the network to obtain the corresponding spatial attention weights and channel attention weights for HSI classification [40]. Roy et al. [41] explored that A2S2K-ResNet combined with spectral attention can extract more cross-channel interaction information and discriminative spectral–spatial features to achieve better classification accuracy. Wavelet-attention CNN divides the feature map into low- and high-frequency components by wavelet transform and introduces an attention mechanism to capture the high-frequency features used for classification [42]. These methods focus attention on the spectral or spatial dimension and achieve the HSI classification with attention-weighted features.

Nevertheless, the approaches mentioned above all focus on a spatial neighborhood or whole spatial range of features rather than at the pixel level. In fact, every HSI pixel contains rich and discriminative features, i.e., pixel-level features. Pixel-level features can be seen as features that describe all spatial pixels of global information. Therefore, making full use of the information contained in all spatial pixel points could improve the classification accuracy of edges and areas that are not easily distinguishable in HSI.

This article explores the implementation of HSI classification task by adaptively selecting filter kernel using pixel-level features, called spatial–spectral adaptive learning with pixelwise filtering (SSALPF). The designed model includes the SSAL module and PF module. In order to obtain all spatial pixel-level features that can describe the global information, the SSAL module first obtains features in a parallel spatial–spectral attention feature extraction manner. Meanwhile, the regularized features act as a guide for the next step of adaptive selection of filter kernel. To provide the PF module with adaptive pixel-level filter kernel, a dictionary composed of filter bases is guided to construct the filter kernel in a linear combined way. In order to obtain the rich and discriminative information contained in all spatial pixel points for classification, the PF module performs filtering operations on HSI using the pixel-level filtering kernel selected above. Therefore, the above operation implements adaptive filtering at the pixel point level and considers the global information contained in all pixel points. The main contributions of this article are summarized as follows.

1) In the SSAL module, a new parallel spatial–spectral feature extraction structure is designed, which fully considers the spatial and spectral information of the original HSI. In addition, the extracted pixel-level features are used as a guide to adaptively select filter kernel from the generated filter dictionary.
2) The PF module implements the filtering of HSI with the filter kernel selected in the SSAL module, making full use of the discriminative information contained in the different pixel points.
3) The proposed network enhances the feature representation of each pixel point and performs well on the three HSI datasets Pavia University (UP), Kennedy Space Center (KSC), and Indian Pines (IP).

The remainder of this article is organized as follows. Section II details the proposed SSALPF architecture for HSI classification. Section III describes the experimental results and the analysis of the results. Section IV conducts the summary.

II. Proposed Method

The proposed SSALPF model mainly includes the SSAL module and the PF module. In order to fully consider the original HSI information and to obtain pixel-level features, SSAL learns spatial–spectral pixel features through a parallel spatial–spectral pixel-level (SSP) feature extraction process. The features are then used as a guide to adaptively select filter kernel from the generated dictionary (GD). The PF module implements the feature representation at the pixel level of the HSI through the filtering operation. Fig. 1 shows the architecture of the proposed method with SSP feature extraction, generate dictionary, and PF. The output of the SSP feature extraction process is Λ. The GD is denoted by D_l. Then, the linear combination of coefficient Λ and dictionary D_l produces F, which is used as the output of the SSAL module. The PF module produces the simple convolution-processed HSI with the filter kernel F to acquire y_{filter}. The original HSI data are represented as $X \in R^{H\times W\times C}$, where H, W, and C represent the height, width, and the number of spectral bands, respectively. First, divide the original HSI into 3-D patch cubes of spatial size $S \times S$, i.e., $X_{\text{pad}} \in R^{S\times S\times C}$. Input 3-D cubes into a network for transformation and network parameter training.

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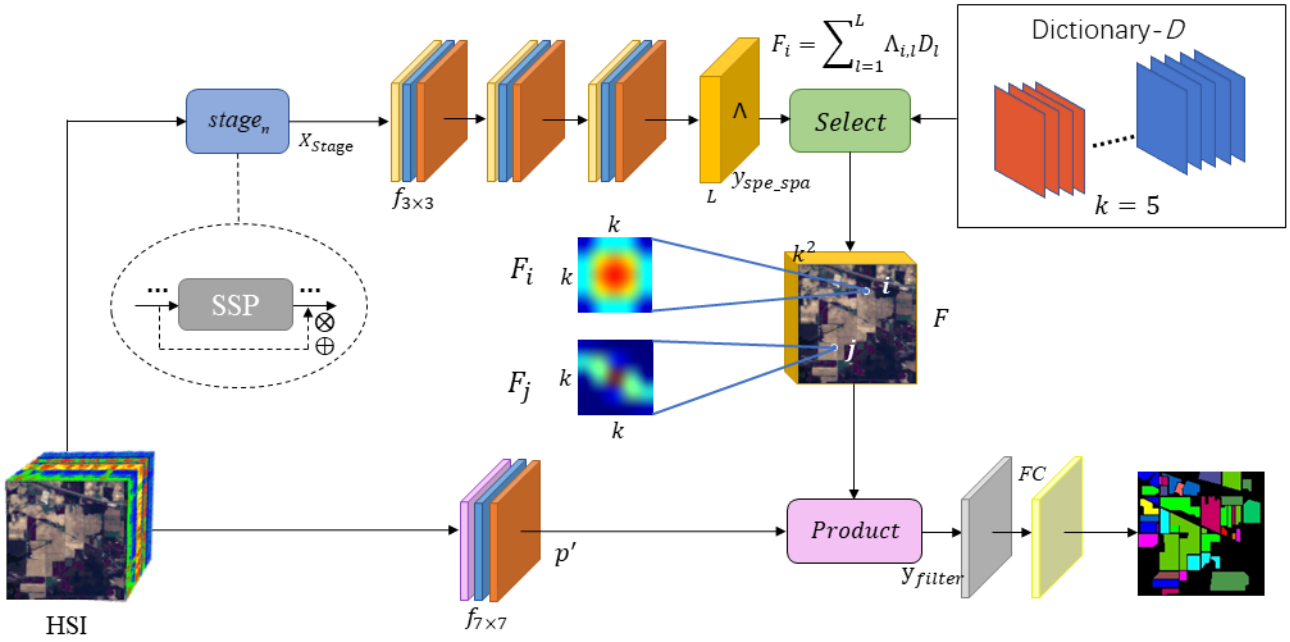


Fig. 1. Structure of the proposed method. First, HSI is used as input to the parallel SSP feature extraction process to learn the discriminative features corresponding to each pixel. The output of this process Λ serves as a guide for adaptive selection of filter kernel from the predefined dictionary D. p' obtained by the simple convolution operation of HSI uses the adaptive filtering kernel F to achieve adaptive filtering at the pixel level. Finally, the FC layer is used for classification.

A. Spatial–Spectral Adaptive Learning

1) SSP Feature Extraction: To enhance the representation of images in HSI classification tasks, the SSAL module is designed to extract joint spatial–spectral discriminative features of each pixel point. Since the attention mechanism has the advantage of focusing on an important region, it is introduced into this module to extract discriminative spatial–spectral features. The structure of the entire module is given as follows. Input X_{pad} into a 2-D convolutional layer with kernel size of 7×7 and output is $X_{7\times7}$.

Then, we capture deep features using n stages, where $n = 4$. To be specific, each stage is stacked with different numbers of bottlenecks. What is more, each bottleneck is made up of an identity mapping with a shortcut connection and a residual mapping. Residual mapping consists of convolutional layers with kernel size of 1×1, 3×3, 1×1, and SSP process. The calculation formula of the proposed SSP can be written as

$$X_{\text{bottleneck},i} = \text{SSP}\left(X_{\text{bottle},i}; \theta_s\right) \tag{1}$$

where $i = 1, \ldots,$ the number of bottlenecks in each stage, and θ_s are trainable parameters in SSP. The number of bottlenecks in each stage is 3, 4, 6, and 3. $X_{\text{bottle},i}$ denotes the input of ith bottleneck and output is $X_{\text{bottleneck},i}$. The hidden mapping is achieved by summing the learned residual mapping and identity mapping.

Therefore, this approach can be of great help in solving the problem that the gradient disappears with increasing network depth and can complement to learn image features at the same time. Then, the identity mapping of $i = 1$ in each stage needs to take downsample for matching output feature map and input feature map in the spatial dimension. The downsample operation is not necessary when $i \geq 2$. After that, the learned hidden mapping is activated by the sigmoid function as the weight value for capturing the spatial–spectral adaptive nonlinear features of the input X_{pad}. Thus, more discriminative information of origin input image can be obtained by weighting the above learned value. After the feature weighting operation, the ReLU activation function is performed as the final output of bottleneck.

The SSP feature extraction process contributes to the acquisition of spatial and spectral features of the image, which is shown in Fig. 2. First, the input feature map is divided into groups along the channel dimension C. Then, each group X_{group} having C/g channels acts as a separate unit and performs the same operations at the same time. The spectral and spatial features of X_{group} are extracted separately and are integrated to obtain the attention-weighted joint spatial–spectral features of the pixel points. To capture long-range cross-channel independent features, spectral attention mechanism can achieve the purpose by recalibration methods [48]. The group feature map X_{group} is executed a global average pooling (GAP) operation by compressing spatial dimension for each channel and maintaining the integrity of channel information

$$X_{\text{GAP}} = \frac{1}{S \times S} \sum_{i,j=1}^{S} X_{\text{group}} \tag{2}$$

where $X_{\text{GAP}} \in R^{1\times1\times C/g}$. To avoid losing some information by direct dimensionality reduction, we keep the dimensionality consistent while learning the features of each channel. The above operation can be implemented by 1×1 convolution with kernel of size k_{gap}. After learning the correlation between each channel and k_{gap} neighboring channels, nonlinear features

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are obtained by the sigmoid activation function and are weighted by X_{group} as output. This process can be expressed as follows:

$$X_{\text{spectral}} = X_{\text{group}} \oplus \left(X_{\text{group}} \otimes \sigma\left(f^{*}_{1\times1}(X_{\text{GAP}})\right)\right) \tag{3}$$

where σ represents the sigmoid function and $f^{*}_{1\times1}$ denotes a 3-D convolution operation with kernel size of $1 \times 1 \times k_{\text{gap}}$, which maintains the channel dimension. $\otimes$ and $\oplus$ denote the multiplication and addition operations by corresponding dimensions, respectively. Then, X_{spectral} is the output feature map of the spectral branch.

Spatial attention is used to capture spatial features by conducting global maximum pooling (GMP) and GAP operations on the feature map in the spatial dimension. The number of feature map X_{group} after GMP and GAP transformations both changes from C/g to 1, integrating both to get the new features of spatial. After that, the spatial attention is generated by convolutional layers with 7×7 kernel and the feature map stitched by channel dimension is reduced to 1. The normalized attention weights are obtained by the sigmoid nonlinear function and then are weighted to the input feature map X_{group} by multiplication operation to complete the recalibration of spatial attention to the original features. The equation for spatial attention feature calibration is given as follows:

$$\begin{aligned} X_{\text{spatial}} = X_{\text{group}} \oplus \big(X_{\text{group}} \otimes \sigma\big(f_{7\times7}(\text{cat}(\text{GMP} \\ \times \left(\text{GN}\left(X_{\text{group}}\right)\right), \text{GAP}\left(\text{GN}\left(X_{\text{group}}\right)\right)))\big)\big) \end{aligned} \tag{4}$$

where GN denotes the GroupNorm option and X_{spatial} represents the output from the spatial branch.

Combine spatial attention and channel attention output feature maps, i.e., X_{spatial} and X_{spectral}, and then, perform the BatchNorm regularization option. The feature map obtained from each group is stitched by channel dimension and the number of channels is restored to be equal to the original input feature maps of C. In addition, channel shuffle is used for regrouping the group convolution feature maps regroup(i), which ensures that the flow of information from different groups can be exchanged

$$\text{regroup}(i) = \text{BN}\left(X_{\text{spectral}} + X_{\text{spatial}}\right)\ i = 1, \ldots, g \tag{5}$$

$$\text{regroup} = \text{cat}(\text{regroup}(i)) \tag{6}$$

$$\text{shu}_{\text{spe_spa}} = \text{shuffle}(\text{regroup}) \tag{7}$$

where BN represents a BatchNorm layer and regroup is obtained by concatenating regroup(i), and then, it outputs $\text{shu}_{\text{spe_spa}}$ after a channel shuffle operation. Therefore, the SSAL module learns the attention-weighted features in the channel and spatial dimensions and achieves feature recalibration. It largely improves the correlation of features in spectral and spatial dimensions and is more beneficial to represent the discriminative features of HSI pixel points.

Finally, three convolutional layers with kernel size of 3×3 reduce the feature map dimension to the number of input channels of the module

$$y_{\text{spe_spa}} = f_{3\times3}\left(f_{3\times3}\left(f_{3\times3}\left(\text{shu}_{\text{spe_spa}}\right)\right)\right) \tag{8}$$

where $f_{3\times3}$ is a convolutional layer and is followed by the BatchNorm layer and the ReLU function. $y_{\text{spe_spa}}$ represents the output of the entire spatial–spectral pixelwise feature extraction process. It is worth noting that $y_{\text{spe_spa}}$ keeps the spatial size consistent and is already normalized, so it can be used to describe the normalized features represented by each pixel in cube X_{pad}.

2) Generate Dictionary: We would like to get more detailed information about the spatial, and since different pixel points may contain rich and different features, the focus should be on spatial information at the pixel level. Normally, we define a general convolution kernel for the weighted average of the whole image without focusing on the pixel points. It has the disadvantage of blurring the features that distinguish details such as edges and textures, which is caused by the fact that all pixel points learn the same convolutional kernel parameters.

Considering that sparse representation enables to compose filter kernel by adaptively selecting filter bases, we also use this approach to generate an adaptive filter kernel for pixel-level HSI classification. Therefore, we first need to generate a dictionary. Then, the filter kernel is learned according to the obtained coefficients' linear combination of filter bases.

We choose the filters based on both effectiveness and complexity. There are many filters that can filter images, such as median filter, wavelet transform, and Gaussian filter. The wavelet transform separates the coefficients of signal and noise according to a threshold, and it is also an ideal tool for time–frequency analysis and processing. However, this also leads to it being very complex. Gaussian filtering is the process of weighted averaging of each pixel point of the whole image and is commonly used in image processing for denoising. Compared to wavelet transform, it is less complex and meets the accuracy requirements. Therefore, we choose the Gaussian filters to form our filter dictionary rather than the kernel function of the wavelet transform. In addition, the filter dictionary consists of difference of Gaussians (DoG) filters in addition to Gaussian filters. Because Gaussian filters have powerful image representation capabilities and DoG filters, as differential representation filters, perform well when dealing with detailed features such as edges and textures. The combination of the two can represent the HSI features more effectively. Therefore, for complexity and effectiveness, our filter dictionary consists of Gaussian filters and DoG filters.

Inspired by [43], the selection of 72 underlying filters ensures the completeness and richness of the dictionary. The dictionary consists of 51 Gaussian filters and 21 DoG filters due to the strong ability of Gaussian filters to represent images and the great advantage of DoG in maintaining detailed information such as edges and image textures. In our network, the dimensionality of the coefficient matrix Λ, i.e., $y_{\text{spe_spa}}$, estimated from the SSP feature extraction process is equal to the number of filter bases, i.e., $L = 72$. Next, the filter kernel F corresponding to $y_{\text{spe_spa}}$ consists of filters selected adaptively from the dictionary for each pixel. Moreover, the filter kernel of size $k \times k$ is compressed to $1 \times k^2$ in order to perform the convolutional option on the original image conveniently, as shown in the following

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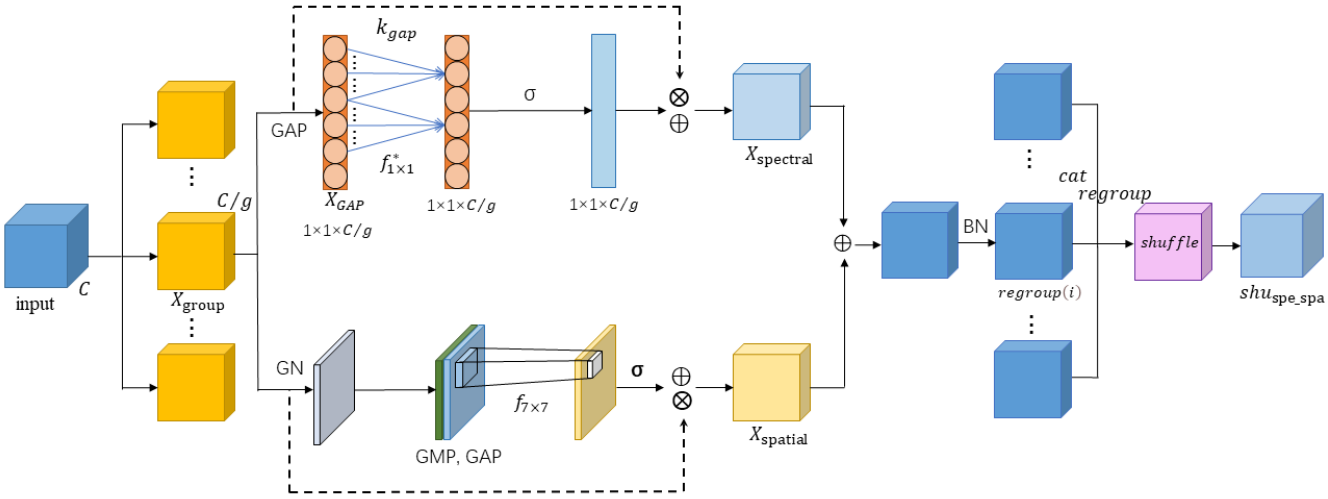


Fig. 2. SSP feature extraction process. To avoid the interaction of high- and low-dimensional features of HSI, the input is first divided along the channel dimension. Then, the channel and spatial attention-weighted features are extracted parallelly and integrated separately. The joint features obtained from each group are shuffled to complete the information exchange between groups and get the final output.

equations:

$$F_i = \sum_{l=1}^{L} \Lambda_{i,l} D_l \tag{9}$$

$$F = \prod_{l=1}^{k\times k} F_i \tag{10}$$

where D_l and F_i represent the lth atom of dictionary D and the filter kernel selected for the pixel at position i in Λ, respectively. Specifically, (9) generates the filter kernel of the ith pixel point, and (10) pairs the shape of the filter kernel of (9) with the pixel points of the HSI to be filtered, facilitating the next step.

B. Pixelwise Filtering

To fully explore the rich and discriminative semantic information contained in all spatial pixels, an adaptive filtering operation is performed on the spatial pixels of the HSI. The general convolution operation uses the same convolution kernel in the whole spatial dimension, so it acquires the features under a specific convolution kernel. Then, it can only obtain imagewise filtering. Unlike the general convolution kernel, our convolution kernel learns the unique features of each pixel in an adaptive way, which is difficult for the general convolution kernel. The learned filter kernel can obtain detailed information about the pixel at the corresponding position, which helps to achieve more accurate classification. Therefore, adaptive filtering is implemented for all pixels in the spatial dimension using the PF kernel. It uses the coefficient matrix Λ obtained earlier as a guide for the filter kernel selection process and then adaptively selects the filter kernel in the dictionary. Then, we need to prepare the feature maps of the HSI to be filtered, i.e., the HSI that is convolved only once. Since the spatial sizes of Λ and the learned filter kernel F and the feature map being filtered are the same, the spatial locations of the pixels correspond. For the filtered feature map, the PF can be performed directly by multiplication.

The process of pixel-level filtering is described in detail next. First, convolution, pooling, and ReLU activation function operations are performed on X_{pad} and output p'. p' achieves dimensionality reduction and retains as much original HSI information as possible. Then, the adaptive representation is achieved by producing $p'_{i,j}$ at position (i, j) in p' with the adaptively selected filter kernel $F_{i,j}$, which can be expressed as follows:

$$y_{\text{filter}} = \sum_{i=1}^{W} \sum_{j=1}^{H} p'_{i,j} \times F_{i,j} \tag{11}$$

where (i, j) denotes the coordinates of the ith row and the jth column pixel of image size $H \times W$ and y_{filter} represents the filtered output. Finally, the pooling and FC layers are utilized to speed up the computation and prevent overfitting and to map the learned feature representation to the corresponding label $\widehat{y}_{i,c}$, respectively. We use the cross-entropy loss function to reflect the prediction effect of each category

$$\text{Loss} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{N_c} y_{i,c} \ln \widehat{y}_{i,c} \tag{12}$$

where N represents the number of all batch samples, N_c is the number of categories included, and $y_{i,c}$ is the real category label.

III. Experimental Results and Analysis

To analyze the effectiveness of our proposed method, we conduct classification experiments on three well-known HSI datasets: UP dataset, KSC dataset, and IP dataset. The performance is evaluated in terms of overall accuracy (OA), average accuracy (AA), and statistical kappa (κ) coefficient. OA refers to the ratio of the number of correctly classified samples to the total number of samples in the test set. AA means the average of category classification accuracy. Kappa represents a strong mutual agreement between the given ground truth and the classification map. To verify the

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effectiveness of our proposed method, we compare traditional machine learning methods and several advanced deep learning algorithms: SVM [11], residual network (ResNet) [44], CBAM [40], ECA-Net [39], A2S2K-ResNet [41], and regularized spectral–spatial global learning (RSSGL) [45]. All experiments are performed on a system with NVIDIA GeForce RTX 3060 12-GB GPU and 16-GB RAM. In order to eliminate the influence of each randomly selected training sample on the results, we conduct ten experiments for each model and calculate the average of the metrics, which are finally presented in the form of mean value ± standard deviation.

A. Experimental Setup

Optimizer and Learning Rate Schedule: To speed up the convergence of the network, we use the stochastic gradient descent (SGD) with momentum optimizer, which sets the optimum learning rate to 0.001 and the momentum value to 0.9. The learning rate is dynamically adjusted when the number of iterations is increased by 15. Then, the batch size of input data is 32 and the training epoch is 300. The space size of HSI cube is set to 13×13. If the accuracy of 20 iterations during the training process no longer improves, we perform an early stop.

B. Datasets Introduction

The UP dataset is acquired by the reflective optics system imaging spectrometer (ROSIS) sensor during a flight campaign over Pavia, Northern Italy. The 103 spectral bands of UP are remained after discarding the meaningless bands. UP is a 610×340 pixels image with a geometric resolution of 1.3 m/pixel. The UP dataset consists of a total 42 776 samples of nine classes. We select 1% labeled samples per class for training, and otherwise, labeled samples are test set.

The KSC dataset is acquired by the NASA Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) instrument over the KSC, Florida, on March 23, 1996. The KSC dataset has wavelengths from 400 to 2500 nm and describes the spatial resolution of 18 m. The spatial of the image contains 512×614 pixels. After removing water absorption and low SNR bands, 176 bands are remained for the analysis. For classification tasks, the KSC ground truth contains 13 classes representing the various land cover types in the scene. As for the UP dataset, 1% labeled samples per class are selected for training and rest for testing.

The IP dataset is gathered by AVIRIS sensor over the IP test site in Northwestern Indiana and consists of 145×145 pixels and 224 spectral reflectance bands in the wavelength range of $0.4–2.5 \times 10^{-6}$ m. The number of covering the region of water absorption bands has been removed is 24 and 200 bands are available. The IP scene contains two-thirds agriculture and one-third forest or other natural perennial vegetation. The ground truth includes 16 classes and it is not all mutually exclusive. We select 2% of the labeled samples as training samples and ensure that the number of samples in each class is at least 5. This approach allows a small sample size of class labeled samples to be used for training.

C. Experimental Design

To fully validate the efficiency of the proposed method, we design several types of experiments. The same hyperparameter settings are used for all comparisons.

1) Comparative Experiments: Tables I–III show the OA, AA, and kappa indicators of the SVM, ResNet, CBAM, ECA-Net, A2S2K-ResNet, RSSGL, and our method on the UP, KSC, and IP datasets, respectively. The best values for each row in Tables I–III are highlighted in bold. The value in parentheses indicates how much lower the compared method is than our method, hence the negative value. In order to see the classification results more clearly, Figs. 3–5 show the visualization results of the classification maps of several methods.

The SVM method divides the features in the sample space into different classes by finding a hyperplane. The ResNet is a deep neural network that uses residual mapping of residual blocks to learn deep features, demonstrating its powerful feature extraction capability. Both traditional and deep learning methods can achieve the HSI classification. The CBAM designs both spatial and spectral attention mechanisms to focus on the identified target objects from the spatial and spectral dimensions, respectively. The ECA-Net proposes an efficient channel attention mechanism that avoids direct dimensionality reduction while improving the interaction capability of cross-channel features. Compared with CBAM and ECA-Net, the A2S2K-ResNet proposes a joint spatial–spectral feature extraction method with selectable convolutional kernel size so that it performs better. The RSSGL model of RSSGL is improved in the learning of adjacent and nonadjacent spectral dependencies and the regularized loss function to extract more discriminative features. By comparison, our approach obtains discriminative features through a parallel joint spatial–spectral feature extraction process and uses the features as coefficients for adaptive learning of the pixel filter kernel. The pixelwise adaptive filtering method can further improve the classification accuracy.

The comparisons show that our method gives better results than the previously mentioned methods on attention mechanisms and the currently best classification methods. The SVM method uses only the original spectral information for classification, while the ResNet network extracts deep features containing the spatial dimension for classification. The CBAM, ECA-Net, and A2S2K-ResNet are all attention mechanism-related methods. The ECA-Net mainly improves the acquisition of channel features in CBAM, which avoids the problem that direct dimensionality reduction cannot fully learn channel features. The A2S2K-ResNet introduces an adaptive convolution kernel based on ECA-Net to obtain more discriminative joint spatial–spectral features. The RSSGL, as a current advanced classification method, learns adjoint and nonadjacent spectral dependencies and can also perform well on the classification task. The proposed method acquires richer and discriminative joint spatial–spectral global pixel features. In addition, the pixelwise filter kernel is adaptively selected for all pixels, making full use of the semantic information contained in each pixel. The PF achieves the classification and yields better results.

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TABLE I
CLASSIFICATION RESULTS OF UP DATASET USING 1% TRAINING SAMPLES WITH DIFFERENT METHODS FOR EACH CATEGORY (%)

Class	SVM	ResNet	CBAM	ECA-Net	A2S2K-ResNet	RSSGL	Proposed
1 (Asphalt)	62.12±1.24	66.73±1.44	61.61±4.22	84.87±1.45	86.50±1.62	91.44±0.58	**94.20±0.73**
2 (Meadows)	73.44±0.52	77.74±0.93	77.52±2.87	93.23±2.08	96.28±0.84	87.93±3.12	**99.38±0.11**
3 (Gravel)	82.38±1.31	91.59±0.25	74.46±3.62	60.01±6.32	76.76±2.48	**92.50±2.10**	85.13±1.17
4 (Trees)	80.26±1.52	93.73±1.01	**99.61±0.02**	98.30±0.04	98.15±0.12	96.41±0.26	96.58±0.45
5 (Paintedmetalsheets)	66.72±1.48	**98.92±0.04**	98.92±0.04	59.98±4.21	95.54±0.51	86.18±2.57	97.54±0.61
6 (Baresoil)	69.35±1.74	90.54±1.33	94.57±0.38	69.87±3.15	97.05±0.35	**100.00±0.00**	98.95±0.02
7 (Bitumen)	52.68±0.89	23.25±4.52	66.81±5.92	75.19±1.62	82.19±3.80	94.15±0.13	**94.40±0.35**
8 (Self − BlockingBricks)	44.13±2.03	48.43±3.88	64.84±3.11	79.27±1.33	77.05±4.92	**99.62±0.01**	83.86±2.04
9 (Shadows)	70.44±1.00	65.54±1.62	77.67±1.53	**93.17±0.21**	62.02±3.66	86.66±1.29	87.37±1.73
OA (%)	70.97 ± 0.11 (−24.19)	72.79 ± 0.13 (−22.37)	75.65 ± 2.32 (−19.51)	85.29 ± 1.04 (−9.87)	91.40 ± 0.47 (−3.76)	92.21 ± 0.14 (−2.95)	**95.16 ± 0.55**
AA (%)	68.72 ± 1.58 (−25.53)	72.94 ± 0.11 (−21.31)	78.92 ± 0.11 (−15.33)	79.32 ± 3.32 (−14.93)	85.73 ± 5.56 (−8.52)	93.60 ± 0.08 (−0.65)	**94.25 ± 1.75**
Kappa (%)	66.98 ± 1.01 (−27.2)	58.95 ± 7.11 (−35.23)	65.42 ± 3.62 (−28.76)	80.33 ± 1.21 (−13.85)	88.53 ± 0.65 (−5.65)	90.42 ± 0.83 (−3.76)	**94.18 ± 0.72**

On the UP dataset, we analyze the results of different studies. The UP dataset mainly consists of asphalt, grass, stone, tree, metal sheet, soil, brick, and shadow. Asphalt includes class 1 (Asphalt) and class 7 (Bitumen), with a classification accuracy of 94.20% and 94.40%, respectively. Our method performs better than other methods on this type. Grass is class 2 (Meadows) and the classification accuracy is 99.38%. Our method has the highest accuracy in this class of recognition and is also more accurate than other methods. Stone is class 3 (Gravel) and the classification accuracy is 85.13%. On this class, the performance of our method is not as good as RSSGL (92.50%), which indicates insufficient sensitivity to this class. Tree is class 4 (Trees) and the classification accuracy is 96.58%. For class 4, our method is not as good as CBAM (99.61%), indicating that CBAM's attention mechanism is more focused on this objective than other methods. Metal sheet is class 5 (painted metal sheets) and the classification accuracy is 97.54%. Although the accuracy is slightly lower than ResNet (98.92%), it also classifies samples as correctly as possible. Soil is class 6 (Bare soil) and the classification accuracy is 98.95%. This class is more accurate compared to the classification results of other classes and slightly less accurate than RSSGL (100.00%). Brick is class 8 (self-blocking bricks) and the classification accuracy is 83.86%. This class achieves the worst accuracy compared to other classes' results. In addition, we guess that the proposed method does not perform well when dealing with movable objects, such as class 3. Shadow is class 9 (Shadows) and the classification accuracy is 87.37%. It is less accurate than ECA-Net (93.17%), probably because the attention mechanism of ECA-Net can identify such objects better. For classes 1, 2, and 7, our method has better recognition for thin boundaries than other methods because it can distinguish details at the pixel level. For classes 3, 6, and 8, RSSGL works well for more dispersed objectives because it obtains global context-dependent information. For classes 4 and 9, some methods based on attention mechanisms achieve better results, probably because they focus more on certain objectives than other methods. For class 5, ResNet achieves the best results, probably because of the small number of misclassified samples. In addition, the classification results for each class can be seen more visually on the classification map (Fig. 3), indicating similar results to other studies.

On the KSC dataset, we analyze the results of different studies. The KSC dataset mainly consists of bush, swamp, woods, and water. In our method, bush is class 1 (Scrub)

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TABLE II
CLASSIFICATION RESULTS OF KSC DATASET USING 1% TRAINING SAMPLES WITH DIFFERENT METHODS FOR EACH CATEGORY (%)

Class	SVM	ResNet	CBAM	ECA-Net	A2S2K-ResNet	RSSGL	Proposed
1 (Scrub)	72.48±0.63	70.88±0.82	86.63±0.71	87.36±0.52	80.53±1.22	98.66±0.01	**99.88±0.02**
2 (Willowswamp)	60.54±0.14	45.81±3.08	54.88±4.25	93.68±0.09	92.35±0.62	91.05±0.13	**94.94±0.72**
3 (Cabbagepalmhammock)	55.63±0.36	58.47±2.54	60.27±2.89	62.18±1.54	78.27±2.89	**86.26±0.25**	69.97±3.16
4 (Cabbagepalm/oakhammock)	22.32±0.48	42.58±2.66	52.87±2.42	**98.82±0.03**	88.62±2.01	89.54±0.42	87.14±1.74
5 (Slashpine)	16.53±1.26	81.30±1.22	72.33±1.02	84.44±0.26	97.71±0.07	95.89±0.06	**98.79±0.13**
6 (Oak/broadleafhammock)	34.69±2.64	70.22±1.04	60.92±0.33	85.73±0.48	80.88±1.66	85.33±1.57	**88.10±0.87**
7 (Hardwoodswamp)	**85.49±0.82**	75.42±0.83	70.26±0.75	74.04±1.22	74.74±2.05	78.39±1.92	79.45±2.53
8 (Graminoidmarsh)	66.18±0.73	61.62±2.73	75.08±0.14	84.62±0.87	82.09±1.57	97.14±0.48	**98.10±0.14**
9 (Spartinamarsh)	82.43±0.32	80.33±1.52	78.49±0.64	92.38±0.42	90.41±1.26	99.25±0.02	**99.36±0.08**
10 (Cattailmarsh)	76.54±1.25	73.18±1.68	83.50±0.29	96.44±0.36	95.43±0.88	**100.00±0.00**	**100.00±0.00**
11 (Saltmarsh)	90.08±0.22	99.58±0.03	95.54±0.05	93.48±0.73	98.84±0.14	99.31±0.07	**100.00±0.00**
12 (Mudflats)	85.89±0.65	87.44±0.72	86.53±0.11	90.92±0.95	92.76±0.34	**100.00±0.00**	94.71±0.83
13 (Wate)	**100.00±0.00**	95.32±0.66	92.24±0.01	98.64±0.08	97.52±0.56	**100.00±0.00**	**100.00±0.00**
OA (%)	70.02 ± 0.58 (−24.66)	68.54 ± 5.98 (−26.14)	82.72 ± 0.55 (−11.96)	88.23 ± 0.50 (−6.45)	91.60 ± 0.48 (−3.08)	93.82 ± 0.22 (−0.86)	**94.68 ± 0.58**
AA (%)	65.84 ± 1.20 (−27.27)	64.66 ± 8.68 (−28.45)	80.62 ± 0.37 (−12.49)	84.22 ± 1.42 (−8.89)	89.40 ± 1.75 (−3.71)	93.00 ± 0.45 (−0.11)	**93.11 ± 0.25**
Kappa (%)	59.54 ± 2.88 (−34.72)	60.45 ± 8.88 (−33.81)	80.25 ± 0.80 (−14.01)	84.27 ± 0.68 (−9.99)	88.83 ± 0.72 (−5.43)	92.94 ± 1.02 (−1.32)	**94.26 ± 0.78**

and the classification accuracy is 99.88%. The class can be almost correctly identified, which is similar to the results in other methods. Swamps include class 2 (Willow swamp), class 7 (Hardwood swamp), class 8 (Graminoid marsh), class 9 (Spartina marsh), class 10 (Cattail marsh), class 11 (Saltmarsh), and class 12 (Mudflats), and their classification accuracy is 94.94%, 79.45%, 98.10%, 99.36%, 100.00%, 100.00%, and 94.71%, respectively. Class 7 is slightly less accurate than the SVM method (85.49%) but higher than the other comparison methods. Class 12 is slightly lower than the RSSGL method (100.00%), which indicates that our method is not sensitive enough in detecting such. Apart from that, our method in other classes has yielded good results. Woods include class 3 (cabbage palm hammock), class 4 (cabbage palm/oak hammock), class 5 (Slashpine), and class 6 (oak/broadleaf hammock) with accuracies of 69.97%, 87.14%, 98.79%, and 88.10%, respectively. The results show that the accuracy is better than other methods in classes 5 and 6, but the accuracy is worse in classes 3 and 4, which we think are misclassified into other classes. Water is class 13 (Wate) and the classification accuracy is 100.00%. The class can be correctly identified, which is similar to the results in other methods. For OA, AA, and kappa, our results are better than other methods on the KSC dataset. In addition, the

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TABLE III
CLASSIFICATION RESULTS OF IP DATASET USING 2% TRAINING SAMPLES WITH DIFFERENT METHODS FOR EACH CATEGORY (%)

Class	SVM	ResNet	CBAM	ECA-Net	A2S2K-ResNet	RSSGL	Proposed
1 (Alfalfa)	42.40±1.53	51.38±2.74	51.42±2.44	45.80±1.92	75.25±0.76	**100.00±0.00**	**100.00±0.00**
2 (Corn − notill)	56.22±1.08	68.05±1.38	54.50±2.86	59.25±2.33	77.92±1.22	86.85±2.16	**95.43±0.38**
3 (Corn − mintill)	54.02±0.72	67.98±1.64	47.10±3.41	53.89±1.46	73.76±1.53	**99.63±0.11**	93.13±1.28
4 (Corn)	50.62±1.32	84.26±0.77	49.77±2.17	71.40±2.11	63.79±1.76	**100.00±0.00**	91.75±0.26
5 (Grass − pasture)	66.71±2.05	80.52±0.42	78.43±1.02	68.26±0.63	**97.63±0.07**	95.98±0.72	97.29±0.03
6 (Grass − trees)	62.58±1.04	82.10±0.56	68.56±0.96	73.28±0.42	88.57±0.52	96.92±0.31	**99.28±0.06**
7 (Grass − pasture − mowed)	60.44±2.13	99.54±0.06	92.95±0.42	78.26±0.71	82.18±1.24	**100.00±0.00**	83.31±2.68
8 (Hay − windowed)	99.54±0.22	97.43±0.09	96.32±0.37	97.18±0.08	99.29±0.07	**100.00±0.00**	97.88±0.32
9 (Oats)	32.43±1.68	43.94±4.36	14.01±4.92	12.82±8.95	52.24±4.13	**100.00±0.00**	73.70±4.63
10 (Soybean − notill)	70.63±0.42	80.61±2.86	63.58±3.16	64.13±3.15	84.90±1.26	91.91±0.52	**93.77±0.61**
11 (Soybean − mintill)	64.51±0.63	60.05±1.44	73.42±1.65	73.24±1.79	86.57±1.18	78.88±2.53	**95.23±0.40**
12 (Soybean − clean)	68.77±0.89	68.41±1.92	58.19±2.44	58.52±2.08	86.63±1.05	**89.16±0.94**	85.32±1.22
13 (Wheats)	92.14±1.02	94.84±0.83	94.25±0.73	90.34±0.54	90.50±0.93	**97.50±0.02**	95.78±0.56
14 (Woods)	80.56±0.04	86.73±0.92	82.80±0.48	84.03±0.63	92.21±0.54	**99.35±0.12**	99.29±0.13
15 (Building − Grass − Trees − Drives)	68.29±0.13	62.35±1.42	61.81±1.26	74.16±1.14	86.13±0.23	**99.74±0.02**	88.04±2.17
16 (Stone − Steel − Towers)	60.44±0.32	70.47±1.86	82.32±1.04	82.33±1.31	75.56±2.66	93.18±0.74	**97.33±0.63**
OA (%)	64.04 ± 0.17 (−30.81)	68.15 ± 1.16 (−26.70)	65.99 ± 0.63 (−28.86)	69.45 ± 1.43 (−25.40)	85.19 ± 1.29 (−9.66)	91.06 ± 1.82 (−3.79)	**94.85 ± 1.52**
AA (%)	59.01 ± 0.21 (−33.76)	70.60 ± 4.36 (−22.17)	62.47 ± 2.79 (−30.30)	66.12 ± 1.51 (−26.65)	80.82 ± 2.58 (−11.95)	**95.57 ± 0.02** (+2.80)	92.77 ± 1.53
Kappa (%)	58.63 ± 1.01 (−35.64)	62.82 ± 1.28 (−31.45)	61.05 ± 0.79 (−33.22)	64.91 ± 1.65 (−29.36)	83.02 ± 1.52 (−11.25)	89.91 ± 1.65 (−4.36)	**94.27 ± 1.69**

classification results for each class can be seen more visually on the classification map (Fig. 4), indicating similar results to other studies.

On the IP dataset, we analyze the results of different studies. The IP dataset mainly consists of flower, crops, grasses, wood, and mixtures. Flower is class 1 (Alfalfa) and the classification accuracy is 100.00%. Compared to other methods, it classifies completely correctly and achieves the best results. Crops include class 2 (Corn-notill), class 3 (Corn-mintill), class 4 (Corn), class 9 (Oats), class 10 (Soybean-notill),

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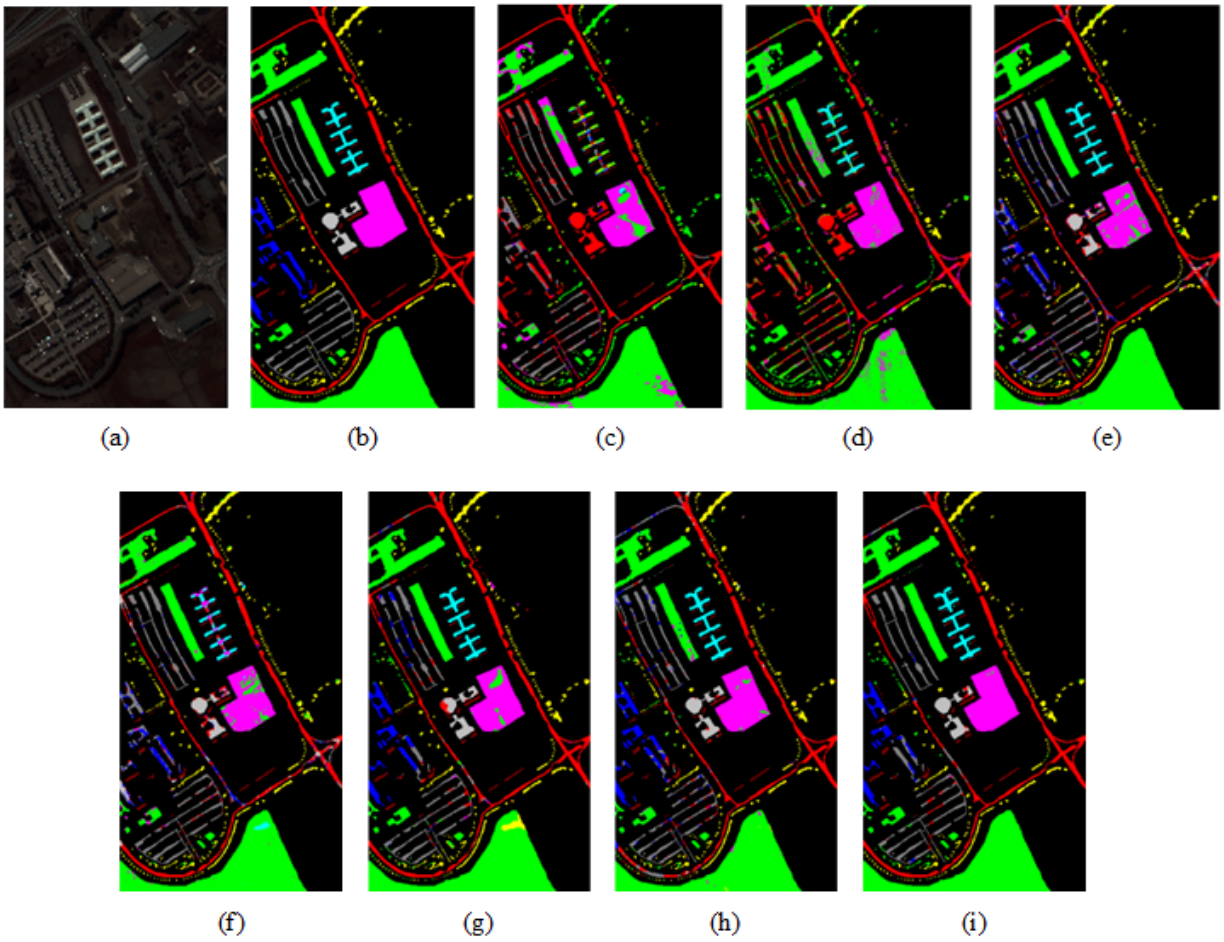


Fig. 3. Classification maps obtained by different methods on the UP dataset. (a) False-color image of three bands. (b) Ground truth. (c) SVM. (d) ResNet. (e) CBAM. (f) ECA-Net. (g) A2S2K-ResNet. (h) RSSGL. (i) Proposed method.

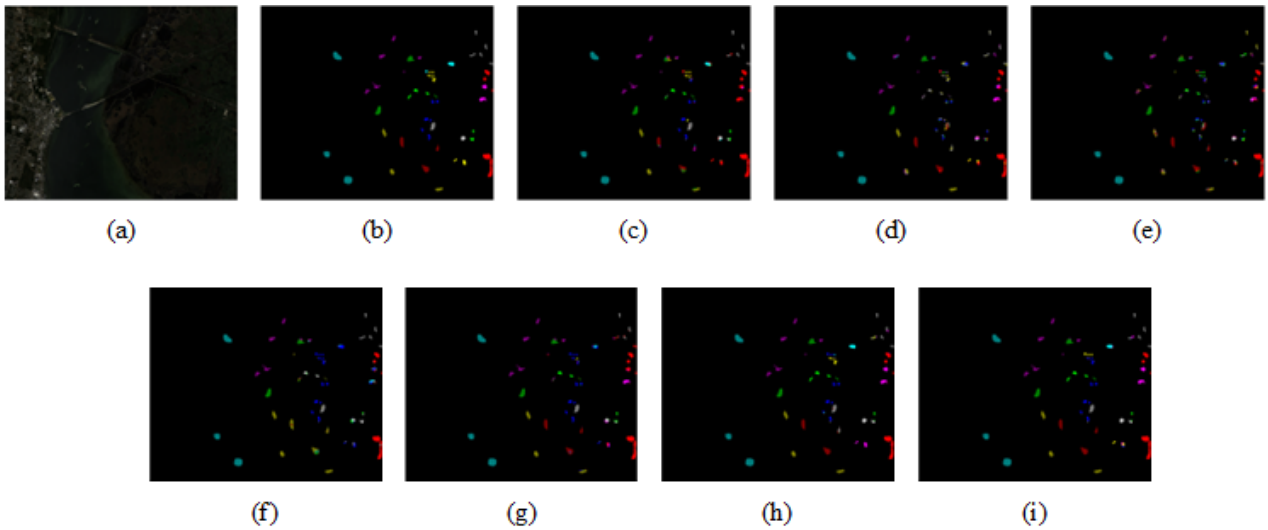


Fig. 4. Classification maps obtained by different methods on the KSC dataset. (a) False-color image of three bands. (b) Ground truth. (c) SVM. (d) ResNet. (e) CBAM. (f) ECA-Net. (g) A2S2K-ResNet. (h) RSSGL. (i) Proposed method.

class 11 (Soybean-mintill), class 12 (Soybean-clean), and class 13 (Wheats), and their classification accuracy is 95.43%, 93.13%, 91.75%, 73.70%, 93.77%, 95.23%, 85.32%, and 95.78%, respectively. Among the classes included in the crop, the overall classification results are more balanced, with some recognition accuracy for different classes. For example, classes 2, 10, and 11 have higher accuracy than RSSGL method (86.85%, 91.91%, and 78.88%, respectively),

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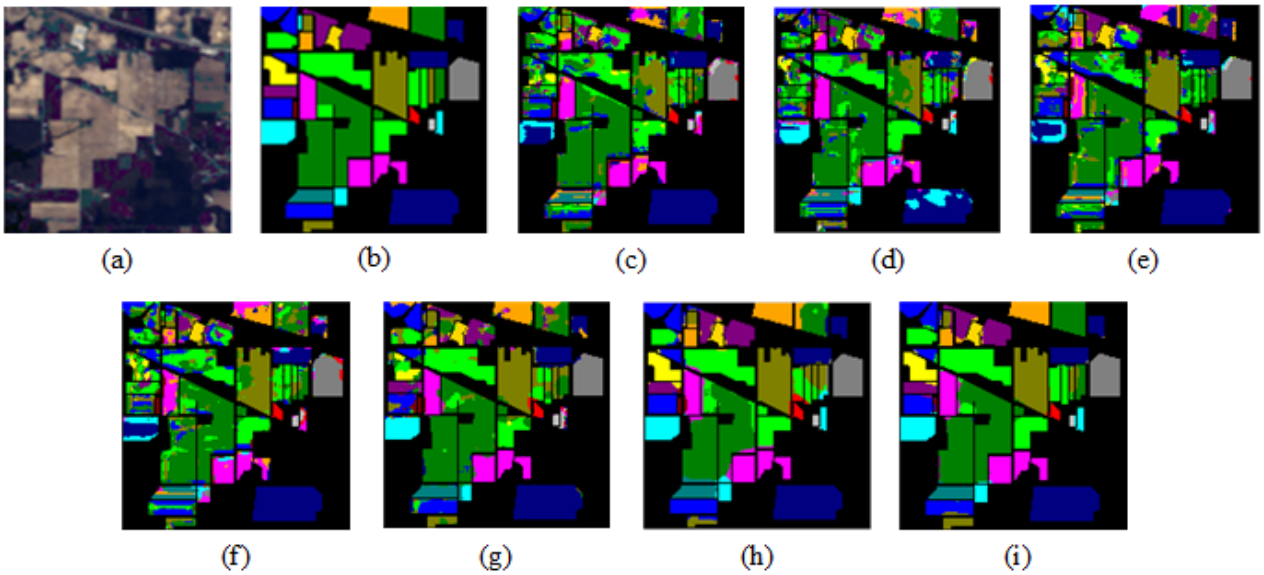


Fig. 5. Classification maps obtained by different methods on IP dataset. (a) False-color image of three-band. (b) Ground Truth. (c) SVM. (d) ResNet. (e) CBAM. (f) ECA-Net. (g) A2S2K-ResNet. (h) RSSGL. (i) Proposed method.

indicating better recognition of these types of crops. For classes 3, 4, 12, and 13, the accuracy is slightly lower than RSSGL (99.63%, 100.00%, 89.16%, and 97.50%, respectively). Class 9 result is significantly lower, indicating poor recognition of oats type of crops. The accuracy of RSSGL is relatively high for most classes, and in particular, some classes can reach 100.00%. Grasses include class 5 (Grass-pasture), class 6 (Grass-trees), class 7 (Grass-pasture-mowed), and class 8 (Hay-windowed), and their classification accuracy is 97.29%, 99.28%, 83.31%, and 97.88%, respectively. In these classes, the overall classification results are not as good as the RSSGL method (95.98%, 96.92%, 100.00%, and 100.00%); in particular, class 7 has a large difference. Wood is class 14 (Woods) and the classification accuracy is 99.29%. It means that the class is basically detected. Mixtures include class 15 (Building-Grass-Trees-Drives) and class 16 (Stone-Steel-Towers), with a classification accuracy of 88.04% and 97.33%, respectively. Our method does not perform well for class 15 and needs to improve the accuracy in the case of multiclass mixing. For OA and kappa, our results are better than other methods on the IP dataset. However, it is lower than the RSSGL method (95.57%) on AA, indicating that there is no advantage in the class average results. In addition, the classification results for each class can be seen more visually on the classification map (Fig. 5), indicating similar results to other studies.

Among the previous methods, our method is higher than the RSSGL with the best performance on the UP dataset for OA (2.95%), AA (0.65%), and kappa (3.76%). From Fig. 3, compared with other methods based on the attention mechanism, we can see that our method shows great advantages in dividing the boundary between the ground objects and the background because of using PF. This is reflected in the classification results of class 6 (Bare soil), which are cleaner than RSSGL. In addition, our method enables each category to be correctly classified with the greatest probability by learning spatial and spectral information adaptively. On the KSC dataset, the OA, AA, and kappa of ours overtop RSSGL by 0.86%, 0.11%, and 1.32%, respectively. Since the number of labeled samples in the KSC dataset is relatively small, it can be seen that our method has obvious competitiveness in dealing with the problem of classification of small samples. The classification maps can also be proved this more intuitively. To be specific, Fig. 4 intuitively sees that class 5 (Slash pine) has more false detections and incorrectly classifies class 8 (Graminoid marsh) into the 1st category (Scrub). On the IP dataset, it can be seen that compared with traditional machine learning SVM (64.04%), the proposed method improves 30.81%, 33.76%, and 35.64% on OA, AA, and kappa, respectively. Then, we compare our method with ResNet, and it is obvious that the proposed method advances 26.70%, 22.17%, and 31.45% on OA, AA, and kappa, respectively. At the same time, ResNet shows better results than SVM. The OA value obtained by A2S2K-ResNet is lower than RSSGL and higher than other methods except our method, though there is still a certain gap with the accuracy of the proposed method. However, RSSGL achieves higher AA than the proposed method by 2.80%. It can also be seen from Fig. 5 that the false detection rate is the highest in class 7 (Grass-pasture-mowed) and class 9 (Oats) and our method mistakenly classifies class 11 (Soybean-mintill) into class 6 (Grass-trees). Besides, the results of class 2 (Corn-notill) are cleaner than RSSGL. After analysis, one possible reason is that the number of prediction errors of most samples is much larger than that of a few samples by RSSGL so that RSSGL is 3.79% and 4.36% lower than our proposed method in OA and kappa values.

In general, our method performs well on UP, KSC, and IP datasets through a large number of comparative experiments with different methods, especially in handling detailed information such as edges. This demonstrates that our proposed PF captures discriminative information from different pixels. In addition, for the classes where the results

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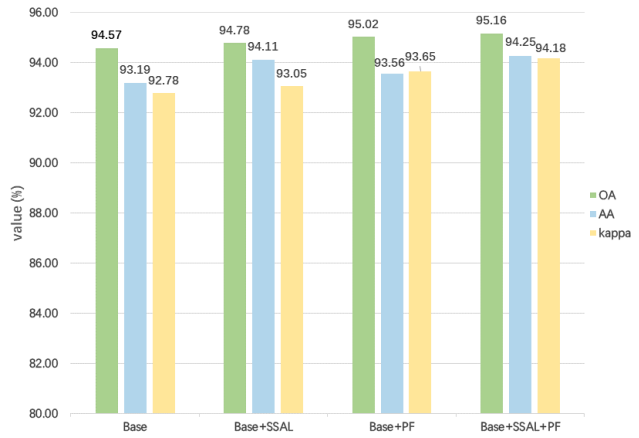


Fig. 6. OA, AA, and kappa indicators of ablation studies on the UP dataset.

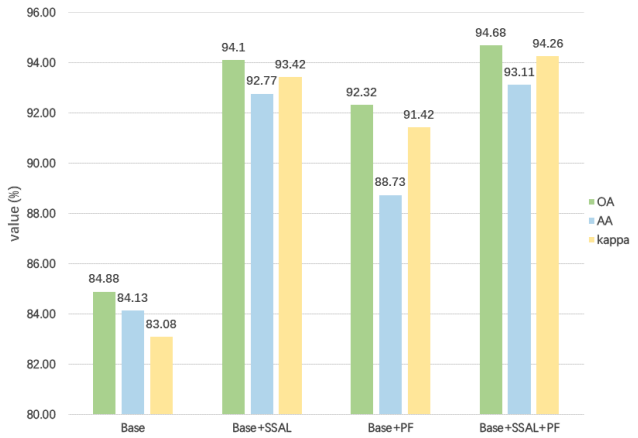


Fig. 7. OA, AA, and kappa indicators of ablation studies on the KSC dataset.

are less satisfactory, we will do further research for these next.

2) Ablation Investigation: Ablation experiments are carried out on the UP, KSC, and IP datasets. To verify the effectiveness of the proposed module, we conduct a series of experiments with baseline.

1) *Base:* ResNet and general filter kernel.
2) *Proposed1:* Base+SSAL.
3) *Proposed2:* Base+PF.
4) *Proposed3:* Base+SSAL+PF.

The Base is composed of the feature vector directly generated by ResNet, a general filter kernel also called F, and the filtered HSI. The dimension of F used as the filter kernel is equal to feature maps generated by ResNet and the filtered HSI processed by the convolution operation. Moreover, this filter kernel dimension is consistent with the adaptive filter kernel. Then, a filtering operation is performed on the filtered HSI feature map using the general convolution kernel, and then, the classification task is implemented. In addition, we add SSP to Base to prove the effectiveness of our proposed SSAL in extracting more complete image features in proposed1. The PF module is combined with Base in proposed2, that is, the general filter kernel in the Base is replaced by an adaptively selected filter kernel from the GD. Specifically, it uses the coefficient matrix obtained from ResNet as a guide to adaptively select pixelwise filter kernel. Therefore, the third experiment compared with Base shows the difference between the adaptive filter kernel in the PF module and the general filter kernel. In proposed3, the feature output from ResNet is replaced by SSAL and serves as a guide to select the pixelwise filter kernel. Then, the adaptive filter kernel replaces the general filter kernel to achieve PF. Therefore, it is designed to demonstrate the effectiveness of SSAL and PF acting on the Base at the same time.

Fig. 6 shows OA, AA, and kappa indicators of the baseline ablation experiment and its variations on the UP dataset. To evaluate the effectiveness of each module, we calculate the OA, AA, and kappa indicators associated with ground truth on the UP dataset. The experimental results indicate that the proposed modules are helpful in improving the classification

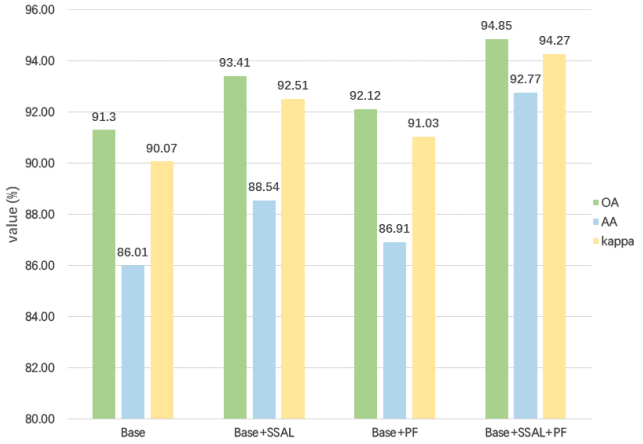


Fig. 8. OA, AA, and kappa indicators of ablation studies on the IP dataset.

accuracy and have some improvements over baseline. Specifically, the OA, AA, and kappa indicators of SSAL module are improved by 0.21%, 0.92%, and 0.27% compared to the Base, respectively. Also, the PF module increases by 0.45%, 0.37%, and 0.87% in OA, AA, and kappa compared with Base, respectively. When both SSAL and PF are applied on Base, it clearly shows that the OA, AA, and kappa promote 0.59%, 1.06%, and 1.40%, respectively. Fig. 7 shows OA, AA, and kappa values of ablation experiments on the KSC dataset and the results indicate that the SSAL has improved by 9.22%, 8.64%, and 10.34% over Base, respectively. What is more, the PF also has an improvement with 7.44%, 4.60%, and 8.34% in OA, AA, and kappa, respectively. The three indicators of the two modules increase by 9.80%, 8.98%, and 11.18%. The ablation experiment results of OA, AA, and kappa on the IP dataset are shown in Fig. 8, in which proposed1 has an improvement of 2.11% over Base in OA and AA, and kappa also increases by 2.53% and 2.44%. For the PF, OA advances 0.82% and the increases in AA are 0.90%, and then, kappa improves by 0.96%. The SSAL and PF modules perform better by 3.55%, 6.76%, and 4.20% when they present together. Therefore, we can prove that the proposed SSAL and PF modules are helpful to improve the classification accuracy.

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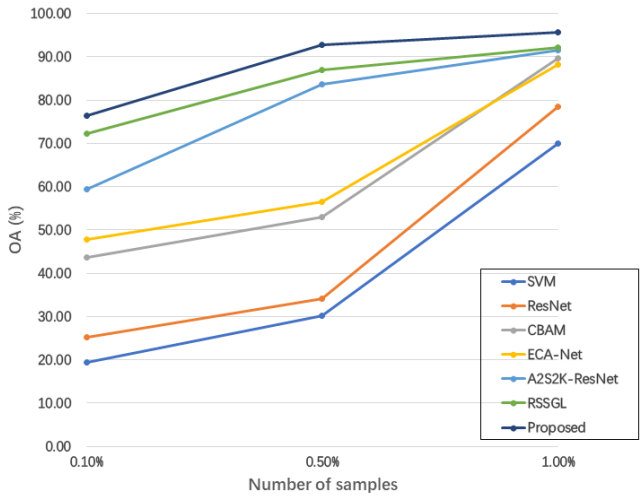


Fig. 9. Effect of different training samples in OA on the UP dataset.

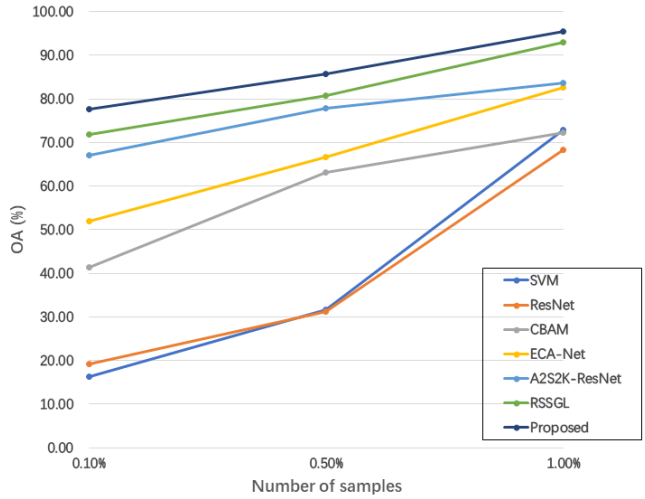


Fig. 10. Effect of different training samples in OA on the KSC dataset.

3) Comparative Experiments of Different Training Samples: In this section, we conduct a series of experiments with different training samples on three datasets and compare our proposed with other methods in the same division. For UP and KSC datasets, we set 1%, 0.5%, and 0.1% of each class as the training samples, and then, the rest are as the test set. The training sets are taken by ten, five, and one samples of each category on the IP dataset, and otherwise, they are testing samples. Fig. 9 shows that the OA (%) values on the UP dataset in which the horizontal axis indicates the number of training samples per category and the vertical axis represents OA (%). As the number of training samples increases, the OA of our method is always higher than that of the other methods on the UP dataset. At the same time, the robustness of our method is reflected in the gentle decreasing trend as the sample decreases. The OA values of different percentages of training samples on KSC and IP datasets are shown in Figs. 10 and 11, which also demonstrate that our method has higher OA and is robust with the same training sample division.

4) Study of the Filter Kernel Size k: For the PF module, we set up a kernel dictionary of size $k \times k$ for building our pixelwise filter F. We conduct experiments on how to reasonably choose k to optimize the classification results on the UP, KSC, and IP datasets; then, the OA, AA, and kappa values of $k = 3$, $k = 5$, and $k = 7$ are shown in Tables IV–VI, respectively.

As we can see from Table IV, the OA reaches the highest value when $k = 5$, k under three values of 3, 5, and 7. OA at $k = 3$ is 0.09% lower than that at $k = 5$ and $k = 7$, which is also 0.88% lower than $k = 5$ on the UP dataset. On the other two datasets, the evaluation indicator OA of $k = 5$ is

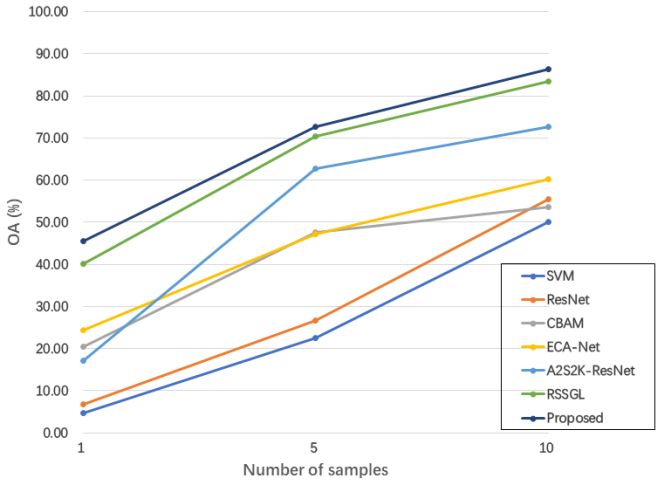


Fig. 11. Effect of different training samples in OA on the IP dataset.

TABLE IV
CLASSIFICATION PERFORMANCE OF KERNEL SIZE ON THE UP DATASET

	OA (%)	AA (%)	kappa (%)
k=3	95.07±0.77	93.59±1.25	94.00±1.01
k=5	**95.16±0.55**	**94.25±1.75**	**94.18±0.72**
k=7	94.28±1.08	92.17±1.01	92.40±1.46

TABLE V
CLASSIFICATION PERFORMANCE OF KERNEL SIZE ON THE KSC DATASET

	OA (%)	AA (%)	kappa (%)
k=3	93.50±1.93	91.98±1.97	92.75±2.15
k=5	**94.68±0.58**	**93.11±0.25**	**94.26±0.78**
k=7	92.32±1.76	89.44±3.67	91.46±1.95

TABLE VI
CLASSIFICATION PERFORMANCE OF KERNEL SIZE ON THE IP DATASET

	OA (%)	AA (%)	kappa (%)
k=3	91.92±1.64	85.96±2.52	90.80±1.86
k=5	**94.85±1.52**	**92.77±1.53**	**94.27±1.69**
k=7	92.20±1.21	87.49±0.60	91.09±1.39

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also higher than $k = 3$ or $k = 7$. In particular, it is increased by 1.18% and 2.36% on the KSC dataset. In addition, the best performance is also obtained by $k = 5$ in both AA and kappa indicators. What is more, we can know that kappa values of $k = 5$ are also higher than other k, which shows that $k = 5$ gets more strong consistency between the classification results and ground truth than $k = 3$ or $k = 7$ on three datasets. In general, to ensure that the dictionary has enough space to cover the filters and to ensure the abundance of the dictionary, $L \gg k^2$. If both large and small object targets exist, we choose a smaller filter kernel, i.e., a smaller value of k, in order to ensure that small objects are not ignored. If the space contains a large number of objects or if the objects are more densely distributed, we should choose a larger filter kernel, i.e., a larger value of k. The advantage of this is that filters can obtain both global information and local information of the objects, which is beneficial for capturing more discriminative features.

IV. Conclusion

In this article, we propose an SSALPF method for the HSI classification task. In order to capture the spectral information contained in hundreds of continuous bands and the spatial information in all pixel points of the space, the SSAL module with spatial attention and spectral attention is applied to extract more discriminative features about each pixel point in a parallel way first. What is more, we conduct PF by using the features of different pixel points to adaptively select the filter from the dictionary. The PF module takes full advantage of the rich and distinguishable features contained in the pixels at different spatial locations. Therefore, the method has a good performance in the HSI classification by filtering each pixel point and considering the detailed information such as image edge and texture. Compared with other methods, our method has competitive classification results on UP, KSC, and IP datasets. The ablation experiments also prove the effectiveness of our proposed module, which can obtain the cross-channel correlation, local neighborhood similarity features, and more diverse features after filtering operation. In addition, the proposed method is also applicable to the field of image denoising and image deblocking.

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